Trader — compiled path

> Compiled from `/home/dnikolic/Documents/Learning/Plan ucenja/paths/Trader`. Canonical sources stay in place.

Phase — 01-market-foundations

Directory: `01-market-foundations/`

Source: `01-market-foundations/01-market-foundations.md`

Market foundations — what you actually trade

Scope

Understand the instruments, liquidity, sessions, and order mechanics that sit *underneath* charts and indicators. Without this layer, execution quality and risk models are guesses.

Positional framing: **trading engineering / execution literacy**, not “predict the tape.”

Core ideas

- **Different products = different constraint surfaces.** Stocks, leveraged FX rollover products, and futures each reshape fills, gaps, financing, halts, and reporting—even when candles superficially rhyme.
- **Price is negotiated in an auction.** Spread, displayed depth, time of session, scheduled shocks change *who* clears your exposure and at what cost.
- **Microstructure leaks into P&L.** Slippage and partial fills belong in expectancy—not a footnote labelled luck.

Areas (with basic elaboration)

Instrument families

- **Single-name equities** — Halts, corporate actions (splits/dividends), exchange microstructure, extended-hours behaviour—expanded in **equities specifics** below.
- **ETFs** — Tracking error; NAV versus last trade premium/discount; liquidity interplay with underlying baskets.
- **Forex spot/roll synthetics via broker products** — Quoting conventions, funding accruals, jurisdictional quirks—expanded in **forex specifics** below.
- **Futures** — Expiry ladders, margins, ticks, curve behaviour distinct from CFD spot metaphors sharing chart skins.
- **CFDs where applicable** — Counterparty disclosures, PRIIPs/leverage safeguards by domicile, financing schedules—study *your* schedule not internet clichés.

Spot versus derivatives framing

Spot vocabulary often masks embedded financing/leverage through broker wrappers—read contract-level behaviour not marketing labels.

Liquidity and friction

Liquidity regimes, dynamic spreads, volatility clustering, sizing interactions with margin calls—foundation for downstream risk modules.

Sessions and actors

Who supplies liquidity versus consumes it shifts intraday—coarse realism beats conspiracy theories.

Execution literacy

Orders, queues, fragmentation where relevant, retail disconnect drill themes—paired later with Broker Validation Engineering for measurement discipline.

Forex specifics — funding, windows, and cross-links to equities sleeves

Operational FX literacy is **ownership of recurring costs**, **event windows**, and **how FX states bleed into equities risk** when you genuinely mix sleeves.

Carry and funding path

- **Carry-trade awareness** — Interest differentials accrue subtly relative to leveraged spot noise while regime shocks can erase months of clipping quickly—carry is contextual, never automatic income.
- **Swap ownership** — Sign and magnitude belong next to journaling rows or blotter summaries; burying swaps poisons expectancy and tax attribution hygiene.
- **Rollover mechanics** — Understand broker cutoff for crediting rolls, correlated spread spikes, ambiguity around “still same UTC day”—misalignment scrambles sequencing stories between gap risk and staged exits.

Policy and scheduled shock

- **Economic-release handling** — Freeze rules, widen-only ladders, size-down ladders decided cold before red folders appear on calendars; tag executions with release-context metadata for later stats hygiene.
- **Central bank posture awareness** — Coarse familiarity with hawkish versus dovish drifts matters because **yield differentials reposition FX** even when discretionary thesis is tactical—prediction optional but hazard signalling mandatory.

Cross-market scaffolding

- **Yield-differential intuition** — Carry math is scaffolding; realised paths often move on repositioning shocks and liquidity droughts—not only tidy interest gaps on a spreadsheet.
- **US dollar index (DXY) as context** — It is an artificial basket—not your traded pair—and correlations with risk assets fracture in stress; use it sparingly as **tone**, not deterministic oracle linkage.

Pair-selection discipline

- **Major versus minor crosses** — Tighter majors often show different liquidity and news beta than minors or exotics; define which families you deliberately trade—and which you forbid when spreads blow out unpredictably versus your journaling history.

Correlation map (lightweight operational)

- Maintain a qualitative **overlap map** bridging FX exposures you hold or hedge implicitly against equity factor tilts—for example exporters versus strong-USD regimes—so macro shocks cannot surprise you stacking invisible correlated bets blindly.

Equities specifics — gaps, corp actions, and session edges

Thin coverage if you purely trade ETFs or indices—but **single-name** reality hits fast without these rails.

Earnings-led gaps

- **Earnings-gap awareness** — After-hours headlines reopen prices far from the prior cash close; declare beforehand whether setups **participate, throttle size, or avoid** earnings windows—not amid headline dopamine.

Corporate actions

- **Splits** — Quantity and nominal price mechanically adjust across vendor charts versus broker rows on mismatched timelines; reconcile so expectancy work does not show fake gains/losses.
- **Dividends** — Ex-div cash timing, withholding, adjustment conventions for charts, and bookkeeping/tax overlays belong explicitly mapped (**verify with advisers**).

Halts & venue stress

- **Market halts** — Pauses scramble intraday liquidation planning; playbook states halt policy (auto-disqualify symbol, escalate manual review tiers, etc.).

Session extensions

- **Premarket / post-market** — Thinner liquidity, wider spreads, different order handling; forbid or selectively permit by playbook—with journaling tags distinguishing **cash vs extended** sessions.

Common pitfalls

- Marketing-grade **FX leverage** versus survivable clustered volatility paths—not the same universe.
- **Chart-only equities** narratives while ignoring corp actions splits dividends halts rewriting live risk mechanically silently.

Basic practice

- Extend your trade memo with forex lines: rollover cutoffs, documented swap polarity, majors/minors policy, macro-release throttle behaviour, discretionary USD-tone context discipline.
- Add equities operational lines: earnings stance, split/div reconciliation checkpoint, halt policy snippet, extended-hours allowed/forbidden tagging.

Outcomes / deliverables

- One-page glossary bridging **instrument → risk surface → session → funding → corp actions**.
- Lightweight **cross-sleeve correlation sketch** tying USD tone plus equity factor tilts you actually hold—honest scribble beats mythical diversification stories.

Phase — 02-broker-platform-germany-operations

Directory: `02-broker-platform-germany-operations/`

Source: **`02-broker-platform-germany-operations/01-broker-platform-and-germany-operations-overview.md`**

Broker / platform / Germany operations overview

Scope

Build the **operational layer** around your trading: custody, statements, reconciliation, archiving, FX and fee visibility, reporting discipline.

Measurement drills (spread/latency/exports/outage rehearsals) belong in **`03-broker-validation-engineering`**; this chapter stays frameworks and workflows—not raw bench tests alone. This is boring infrastructure—until audits, outages, broker changes, or tax prep prove it mandatory.

Anything below marked **Germany** is **study scaffolding** for workflows you confirm with authoritative sources and a **tax professional (Steuerberater)**—not legal facts frozen in Markdown.

Core ideas

- **Your archive is inventory.** If broker UI changes tomorrow, immutable exports dated by tax year survive.
- **Fees are expectancy.** Commissions + spread + swaps + dividend handling + FX markup belong in expectancy math and reviews.
- **Regulation sets guardrails.** Product access (retail CFD rules, PRIIPs, leverage caps, investor warnings) differs by geography—verify continuously.

Areas (with basic elaboration)

Selection & tooling

- **Broker selection framework** — product coverage, segregation/protection disclosures, outages history, stability of exports, jurisdiction and compensation schemes (study official docs).
- **Platform comparison** — charting vs execution fidelity; journaling hooks; portability of history; scripting if you automate journaling.
- **Execution quality cues** — requotes/rejects routing transparency; outage playbooks—how you flatten or hedge when platform fails.

Regulation basics (themes to verify)

- **Regulation primer** — who supervises whom, segregated accounts language, suitability/knowledge tests where applicable—not one paragraph “answer.”

Germany-oriented operations (confirm with advisor)

- **Tax overview thematic** — when trading income can touch Abgaben in ways that deserve planning; thresholds and classification are individual—avoid fixed numbers unless copied from dated official guidance tied to YOUR case.
- **Private vs Gewerblich / commercial patterns** — only an advisor maps habituality, intentions, ancillary income, bookkeeping depth.
- **Reporting obligations checklist** — which broker exports satisfy which filing lines historically; **annual refresh** mandatory.

Workflows

- **Trade export workflow** — schedule; checksum or hash optional; filenames with date range & broker id.
- **Documentation workflow** — ticket → blotter → reconciliation to monthly statement deltas.
- **Broker statements vs internal ledger** — treat mismatches as bugs until explained (corporate action, FX, delayed settlement).
- **Trade archive ownership** — who stores what, redundancy (offline copy), naming convention.
- **Compliance ownership calendar** — what you owe and when—even if outcome is “not applicable.”

Structure & currency

- **Account structure** — single vs multicurrency wallets; segregation of speculative vs investing journals if legally distinct in your posture (advisor).
- **Conversion awareness** — functional currency bookkeeping vs broker display currency traps.
- **Cost structure ledger** — every debit line categorized (execution, idle cash drag, borrowing, dividends tax withholding).

Common pitfalls

- Trusting screenshots without **statement reconciliation**.
- Migrating brokers without frozen **prior-year artefacts**.
- Using third-party calculators for DE tax specifics without validating against **annual Finanzamt forms** counsel uses.

Basic practice

- Run a **reconciliation dry run** spanning a full brokerage statement cycle: scheduled exports → internal blotter match → annotate FX rows and fee categories until gaps clear or are explained.
- Write a **broker switch playbook** listing artefacts to migrate before goodbye (fills, dividends, withholding, FX).

Disclaimer

Treat tax and residency-specific rules as **living research** aided by counsel; keep a dated log when assumptions change.

Phase — 03-broker-validation-engineering

Directory: `03-broker-validation-engineering/`

Source: **`03-broker-validation-engineering/01-broker-validation-engineering.md`**

Broker validation engineering — practise the operating layer

Scope

Operational quality is not “which broker has the best banner ad.” It is **measurable**: spreads you pay, latency you feel under load, whether statements reconcile, exports parse, and whether drills exist before real outages.

Pairs naturally with `02-broker-platform-germany-operations` (selection, tax workflow themes). This unit is **measurement + rehearsal** on the stack you already chose.

Core ideas

- **Expectancy hides in execution plumbing.** Quiet spread widening and occasional bad prints compound like any other recurring cost—log them deliberately.
- **Drills outperform hope.** Simulate outage and export recovery while calm; rehearsing hot loses money and sleep.

Areas (with basic elaboration)

Spread measurement

- Sample bid–ask snapshots around your actual trade windows—not only midday calm. Log instrument, session slice, weekday vs rollover proximity.
- Separate **broker quoted spread** from **effective spread** paid on marketables when you deliberately cross.

Latency measurement

- Clock click-to-acknowledgement subjective tiers (platform UI responsiveness) versus objective order round-trip anecdotes where platform exposes timestamps.
- Note behaviour under news spikes—not only idle charts.

Execution comparison

- Compare execution artefacts across sessions and volatility buckets—rejects, partials, slippage versus plan—not only “feel.” Tag causes where you can distinguish platform vs liquidity vs impulse.

Statement validation

- Monthly reconciliation drills: blotter ↔ PDF/CSV statements; unmatched rows must close with an explanation recorded (corp action, FX rounding, odd fees, splits).

Platform outage drills

- Written path to flatten reduce or reconcile when UI fails mid-bracket; redundant path on second device where policy allows it; acknowledge phone-queue reality if API-only fails.

Export validation

- After broker UI upgrades, reopen your usual CSV columns and ingestion rules—a silent column rename breaks archives and adviser handoffs downstream.

Mobile vs desktop workflow

- Declare which workflows are forbidden on mobile (complex brackets, scaling ladders) versus allowed only (flatten, cancel all). Divergence breeds silent improvisation.

Multi-monitor ergonomics

- Stable screen layout anchors blotter latency clock news calendar journaling pane—reduces mis-clicks fatigued hours.

Hotkey safety

- Keep a numbered hotkey map; avoid ambiguous combos that flip market vs limit under stress; rehearse reversing a mistaken keystroke calmly.

Common pitfalls

- Trusting promotional “spread from” screenshots instead of journaling **your** prints across weeks.

Basic practice

- Broker **scorecard artefact**: three columns labelled **measure | method | acceptable band** spanning spread samples latency notes export parse success outage drill date.

Outcomes / deliverables

One living **broker validation scorecard** plus one **outage drill checklist** exercised at least once on paper before you rely on adrenaline.

Phase — 04-risk-engineering

Directory: `04-risk-engineering/`

Source: `04-risk-engineering/01-risk-engineering.md`

Risk engineering — survival budgets

Scope

Treat risk as **engineering**: explicit budgets, escalation paths, and circuit breakers. Edge expressed without risk governance is statistically invisible—ruin dominates long before "edge" manifests.

Core ideas

- **Risk is path-dependent.** Correct small risks repeated under leverage can accelerate loss streaks exponentially.
- **Limits pre-commit you.** Decide cold-state rules hot-state will abuse if left implicit.
- **Position size is compression.** Compress exposure until variance of outcomes becomes personally tolerable and journal-honesty improves.

Areas (with basic elaboration)

Trade-level

- **Position sizing primitives** — stop distance × contract multiplier / nominal vs account fraction; unify units you actually trade.
- **Risk per trade** — max loss expressed in R or currency AFTER including gap/slippage cushion—never naive "risk to stop midpoint only."

Account-level

- **Account risk model** — aggregate simultaneous risk when correlated positions share one macro shock.
- **Max drawdown rules** — when you stop trading systematically vs scale down—not vague "feel bad."

Throttle ladders

- **Daily / weekly / monthly loss caps** — different severities trigger stand-down lengths and review gates.
- **Leverage governance** — hard multiplier caps irrespective of broker offers; escalate only after audited review —not after wins.

Exit philosophy

- **Stop-loss vs invalidation** — mechanical stop versus structural thesis break; clarity prevents moving goalposts silently.

Concentration & correlation

- **Concentration** — thematic overlap (seven "different" EURUSD overlays are one macro bet).
- **Exposure ownership ledger** — who holds what directional delta by sleeve.
- **Portfolio correlation hooks** — co-movement regimes shift effective risk multipliers.

Advanced awareness (study, rarely trade raw)

- **Vol-conditioned sizing hooks** — scale risk down when volatility distribution shifts—document rule not gut.
- **Kelly awareness** — full Kelly lethal for estimation error; fractional Kelly discourse only after stable measurement window.
- **Ruin intuition** — many small favourable edges lose to absorbing barrier (bankruptcy threshold) faster than intuition suggests—simulate conceptually via literature.

Common pitfalls

- Resetting caps after revenge session "because I'm sharper now."

- Underestimating **gap risk** sleeping through illiquid exposures.
- Counting winnings as justification to loosen rules **during** winning streak—the classic pre-drawdown euphoria.

Basic practice

- Write **risk sheet**: max contracts/shares at max stop distance ⇄ nominal loss ⇄ % account with slippage add-on clause.
- Predefine three **circuit breakers**: daily loss halt, streak halt (N losses), drawdown throttle from peak equity—with cool-off lengths.

Outcomes / deliverables

- Documented risk sheet wired to sizing.
- Printed **circuit breaker policy** taped near workspace or referenced in journaling template.

Phase — 05-risk-reward-engineering

Directory: `05-risk-reward-engineering/`

Source: `05-risk-reward-engineering/01-risk-reward-and-expectancy-engineering.md`

Risk : reward engineering — measuring quality not hype

Scope

Separate **statistical merit** from **storytelling**: risk multiples, payoff asymmetry, expectancy, distribution shape. Winning trades neither prove excellence nor disprove a broken process.

Core ideas

- **Expectancy is an average over trades**—silent about streak length and pathology of worst outcomes.
- **Win rate is trade space** versus **payoff asymmetry**; extremes can coexist in sustainable systems—or mask overfitting briefly.
- **Quality metrics must align with how you Journal & tag** setups; orphaned trades poison statistics.

Areas (with basic elaboration)

Ratios

- **Risk : reward (R:R)** — planned multiples vs empirical distribution of realised multiples (often uglier).

Expectancy

- **Positive expectancy formulation** — (win% × avg win) − (loss% × avg loss), extended with costs and partial fills—not chart fantasy.
- **Expectancy sensitivities** — how fragile expectancy is when win rate dips 5 points or payoff skew compresses—sanity modelling.

Trade-offs

- **Win-rate ↔ payoff asymmetry** — accepting lower win-rate for asymmetric payoffs shifts psychological load; plan psyche budget too.

Diagnostics

- **Profit factor** — gross wins / gross losses; brittle with small samples; useful trend warning when journaling deep.
- **Edge measurement pipelines** — compare tagged subsets vs naive baseline subsets (avoid self-serving slicing).
- **Payoff distributions** — skew concentration: many small winners / rare large losers or inverse—each dictates risk engineering.

Averages

- **Average win / average loss sanity** — check fat tails—not only means.

Sustainability

- **System sustainability narratives** — can drawdown amplitude still exist within capital runway & psychology budget?
- **Variance awareness** — confidence intervals widen with path length; humility tables matter more than headline expectancy.

Common pitfalls

- Declaring expectancy after **<30–50 homogeneous trades** tagged identically—but even then remain sceptical depending on rarity.
- Excluding worst trades as **“mistakes Outlier”** when pattern recurringly signals playbook flaw versus discipline flaw—disambiguate ruthlessly.

Basic practice

- Build rolling stats view: expectancy, payoff ratio by **single playbook**, streak lengths, longest adverse excursion aggregates (even approximate).
- Every month annotate **why metrics moved** versus random noise hypotheses.

Outcomes / deliverables

Rolling stats artefact keyed to **labeled trade sets**.

Phase — 06-trade-system-development

Directory: `06-trade-system-development/`

Source: `06-trade-system-development/01-trade-system-development.md`

Trade system development — repeatable playbooks

Scope

Industrialise discretionary or hybrid intuition into **finite rules**: explicit premises, confirmations, exits, invalidation. Goal: reproduce conditions under scrutiny—not inspirational bullet lists drifting weekly.

Distinction clarified: **System** ≠ infallibility; it's **explicit falsifiability** + logging.

Areas (with basic elaboration)

Thinking framework

- **Hypothesis formulation** — what would prove this wrong quickly vs slowly?
- **Market premise ownership** — why imbalance should exist transiently—not eternal truth claims.

Rule precision

- **Entry criteria** — objective triggers + discretionary overlay zones marked explicitly if hybrid.
- **Exit criteria** — partial scale rules, trailers, timed exits—all pre-decided tiers.
- **Invalidation triggers** separate from nominal stop semantics—structural falsification pathway.

Context layers

- **Confirmation hierarchy** — structure → momentum proxy → contextual filter ordering (document order effect).
- **Trend operating framework** — definition of regime you trade for trend plays (can be heuristic not indicator dogma).

Adjacent styles

- **Mean reversion awareness** — where fading works until structural breaks vaporise edge historically for style X.
- **Breakout playbook awareness** — false vs continuation clusters; contextual filters distinguishing both.

Structure & liquidity thinking

- **Liquidity constructs** referencing your style—not generic memes—these scaffold where stops cluster logically *for you* given documented premises.

Volatility & constraints

- **Volatility regimes** altering threshold distances or playbook activation flags.
- **System constraints catalogue** — hours, news freezes, blackout periods, correlated market dependencies.

Governance

- **Rule RACI** — Responsible / Approver roles if co-developing—or future-self approval windows (cooldown before patch).
- **Consistency auditing schedule** — detect silent drift (changing definitions without versioning).

Common pitfalls

- **Retrofitting narrative** onto random winners—backward induction bias.
- **Indicator proliferation**: each rule needs removal test—would system lose falsifiability entropy if removed?
- **Blending incompatible premises** trend + mean revert without session tag separation—expect ghost statistics.

Basic practice

- Write **two-page playbook** for one setup only: premise, filters, triggers, sizing, exits, anti-patterns checklist.
- Add **experiment backlog**: tests that could invalidate premise if failed (replay + forward paper).

Phase — 07-market-execution-engineering

Directory: `07-market-execution-engineering/`

Source: `07-market-execution-engineering/01-market-execution-engineering.md`

Market execution engineering — turning rules into fills

Scope

Close the gap between playbook text and brokerage reality: deliberate order choreography, timing, partial fills, event handling. **Ideas plus bad prints** produce negative expectancy even with theoretically sound triggers.

Core ideas

- **Instrument × venue × volatility** determines whether your order type behaves like training videos or battlefield chaos.
- **Simplicity hides failure modes.** A market click is mentally easy but structurally explosive in thin books during spikes.
- **Execution risk is orthogonal to directional thesis** sometimes—you can be thesis-right and mechanically shredded.

Areas (with basic elaboration)

Order mechanics

- **Limit vs market semantics** — control of price versus certainty of placement; toxicity during news spikes.
- **Stop vs stop-limit** — gap risk semantics (trigger vs guaranteed fill)—platform-specific; read disclosures.
- **Partial fills sequencing** — working remainder exposure; cancelling incomplete stacks risk leaving accidental naked bias.

Cost awareness

- **Slippage ownership** — classify systematic vs episodic spikes; annotate major news vs structural thinness independently.
- **Spread awareness overlays** — when spread widens mechanically invalidates discretionary edge—you need pre-code behaviour (stand down vs reduce size tier).

Timing & ladders

- **Liquidity probing intuition** — not prediction; avoidance of habitual entries into known thin windows unless sized for adversity.
- **Execution timing regimes** — open auction behaviour vs mid-session vs close imbalances (instrument-specific study).
- **Order ladder (“working”) framing** scaling in/out without improvisation drift—planned grid vs emotional averaging.

Context

- **Session timing interplay** — roll windows, midday decay, lunchtime gaps.
- **News / event volatility handlers** — pre/post freezes, widen-only tiers, downgrade size multipliers—all written cold.

Resilience

- **Execution failures** — disconnect mid-order, rejected modification, stale quote bursts; scripted recoveries minimise improvisation under cortisol spike.
- **Gap risk awareness** — overnight / weekend exposures; weekend FX gap mental model differs from equities gaps structurally depending on venue.

Common pitfalls

- Chasing breakout with escalating market clips during spread explosion.
- Blaming broker for predictable **stop cascading** regimes you willingly entered leveraged.

Basic practice

- Build **single-page execution playbook** mapping each setup tier → primary order path → fallback → abort conditions.
- Log **five slippage episodes** tagging cause class: spread widening, volatility, thin book, rushed click, conflicting bracket child order.

Outcomes / deliverables

Canonical execution playbook artefact revisable quarterly.

Phase — 08-backtesting-engineering

Directory: `08-backtesting-engineering/`

Source: `08-backtesting-engineering/01-backtesting-engineering.md`

Backtesting engineering — evidence with humility

Scope

Backtesting is inference under adversarial realism constraints: data limits, hindsight bias temptation, unrealistic fills. Goal is **risk-controlled discovery** feeding deployment gates—not shrine worshipping equity curves.

Methodological stance: every backtest emits an **explicit assumption ledger** listing optimistic shortcuts—you cannot eliminate bias, only expose it.

Areas (with basic elaboration)

Method spectrum

- **Manual backtests** discipline thinking before tooling automates sloppy shortcuts.
- **Replay systems** aligning bar / tick fidelity with realism claims—coarse aggregates hide microstructure artefacts.

Statistical stewardship

- **Sample size stewardship** rare setup paradox: long frequency requires years; impatient scaling invites noise trading.
- **Survivorship bias** — equities universes shedding dead names; leveraged index products rebalance artifacts; understand dataset lineage.
- **Look-ahead bias vectors** corporate action adjustments, peeking peaks of future volatility estimators unknowingly anchored forward.

Model risk

- **Overfitting vigilance** — parameter sweeps yielding gorgeous in-sample arcs; humility via holdouts / alternate eras if data supports.
- **Robustness probes** perturb parameters mildly—does edge evaporate twitchily? Brittle optimisation signals narrative curve fitting.

Process discipline

- **Parameter freeze windows** experimentation vs production segregation.
- **Walk-forward scaffolding** pragmatic simplified variant: alternating windows when data depth exists—still interpret cautiously.

Evidence stages

- **Out-of-sample holdouts** when statistically meaningful separation possible—often retail sample starved acknowledge limits honestly.
- **Edge verification narratives** juxtaposed with naive benchmarks (buy-hold analogous where logical).
- **Benchmark ownership** evolves—changing macro regimes shift passive baselines; refresh comparisons.

Common pitfalls

- **Frictionless fills** at bar highs/lows magically.
- **Implicit future information** leaking through indicator normalisation referencing full series.
- **Cherry-picking start dates** aligning with favourable macro window.

Basic practice

- Run intentionally **stress-broken** replicate: widen assumed slippage + spread + pessimistic latency fudge—does skeletal edge persist?
- Document **five optimism risks** appended to study README before sharing or scaling.

Outcomes / deliverables

Study bundle + **explicit assumption ledger** listing realism shortcuts.

Phase — 09-replay-simulation-engineering

Directory: `09-replay-simulation-engineering/`

Source: `09-replay-simulation-engineering/01-replay-and-simulation-engineering.md`

Replay / simulation engineering — practice runway

Scope

Cheap, disciplined repetitions before capital risk saturates cognition. Bridge from abstract backtest artefacts to embodied decision latency comparable to forward decision windows.

Abbreviated prudent progression reminding purpose only:

```
backtest (structural hypotheses)
→ paper / sim path (execution + journaling discipline)
→ small-size live probing (risk governance real)
→ controlled scale monitoring
```

Core ideas

- **Practice architecture shapes habits** more than affirmation quotes.
- **Simulation fidelity discrepancies compound** stealthily—document where sim diverges early.
- Emotional rehearsal separates **panic loops** partially—never fully substitutes live variance.

Areas (with basic elaboration)

Modalities

- **Replay workflows** — cold scripted sessions logging decisions BEFORE reveal of future bars—not peeking cheats.
- **Simulator discipline parity** bracket behaviour, sizing constraints, disallow fantasy partial fills unrealistic relative to broker.
- **Paper trading integrity** policing unrealistic fills latency assumptions; journaling identical fields as live—not lazy shorthand.

Drills

- **Execution repetition drills** micro-cycle same setup family until clicking path automatic without improvisation.
- **Scenario tabletops** scripted shock injects connectivity loss midway bracket adjust—verbal rehearsal OK first.
- **Edge rehearsal vs emotional rehearsal** distinguishing mechanical muscle memory from adrenaline exposure.

Transition hooks

- **Execution timing rehearsals** aligning with session realism you anticipate live—not midnight casual randomness divorced from physiology.

Reviews

- **Replay reviews** mirroring journaling schema—grade process not peeking hindsight P&L of alternate paths obsessively unhealthy—balance.

Common pitfalls

- Treating indefinite paper indefinitely—**risk governance never activates** cognition fully.
- **Cheating hindsight** unconsciously by scrubbing inconvenient branches during replay narration.

Basic practice

- Define **explicit promotion gates** bridging sim → smallest live tier (sample window + review KPI + drawdown leash).
- Log **five sim vs live divergence notes** capturing fill realism deltas after small-size starts.

Phase — 10-journaling-systems

Directory: `10-journaling-systems/`

Source: `10-journaling-systems/01-journaling-systems.md`

Journaling systems — no journal, no system

Scope

Turn memory into **queryable artefacts** bridging psychology → process → evolution. A modest schema you actually maintain beats ornate templates abandoned after novelty fades.

Incomplete or inconsistently tagged logs quietly corrupt later statistics—treat tagging like schema migration discipline.

Core ideas

- **Garbage-in statistics are worse than ignorance**—they feel scientific while lying.
- **Pre-defined fields** reduce post-hoc myth-making about discipline and intent.
- A journal unifies **risk, playbook, and psychological state**; splitting across incompatible tools fractures analysis later.

Areas (with basic elaboration)

Phases

- **Pre-trade entries** capture hypothesis, mapped invalidations, allowable adverse excursion wording, sizing rationale anchored to risk sheet references.
- **Intra-trade notes** only when material state changes—not compulsive twitch logging unless your style mandates it explicitly.
- **Post-trade reconstruction** sequences intent → actions → fills → deviations with timestamps where platform allows.

Artefacts

- **Screenshots cadence** — minimal timestamped anchors proving context, not unbounded hoarding draining review energy.
- **Emotional tagging** — simple anchored scale examples so “fine” isn’t meaningless noise.

Taxonomy backbone

- **Setup / playbook tags** power downstream stats — prevent synonym creep via periodic taxonomy audit (collapse duplicate labels mechanically).
- **Mistake tags** start coarse categorical families until data justifies finer splits—premature granularity fragments samples.

Deviations & multiples

- **Deviation-from-plan** tagging distinguishes improvisation from unavoidable microstructure hits—different remediation paths downstream.
- **Expectancy rollup hooks** plan early—even if dashboards come later—to avoid rewriting history fields retroactively painfully.

Metrics glue

- **R-multiples** logged with explicit definition alignment (planned vs realised if both tracked—never silently mix streams).

Bridging rituals

- **Review ritual linkage** journaling fields must funnel into backlog + versioned playbook patch proposals—not emotional vent silo devoided action accountability.

Common pitfalls

- Expanding templates **before habitual honest usage** slows adoption—iterate minimally only when an analytic question is blocked demonstrably.
- Editing or omitting inconvenient trades sabotages expectancy stats and retrospective honesty.

Basic practice

Start a **minimal field set** documenting hypothesis, playbook tag, planned risk units, adherence boolean, qualitative emotional tag, deviation tag line—expand reluctantly evidencing analytic necessity surfaced reviews.

Phase — 11-trade-review-engineering

Directory: `11-trade-review-engineering/`

Source: `11-trade-review-engineering/01-trade-review-engineering.md`

Trade review engineering — where deliberate improvement compounds

Scope

Institutionalise **graded reviews**: separate process quality from nominal P&L noise. Serious operators ask first whether protocols executed—money second.

Quote discipline directionally aligned with professional attitude:

> Lose money by rule adherence sometimes signals integrity; chaotic wins degrade future discipline capital.

Institutional scepticism warns: analysing only **\$ loss** overlooks subtle **\$ won badly** underwriting future calamity sizing confidence.

Core ideas

- **Grade tracks orthogonal dimensions** concurrently: thesis, timing, sizing, adherence, outcome (outcome informs but must not tyrannically override short windows).
- **Pattern detection requires categorisation taxonomy** matured slowly—don't micro-slice prematurely.

Areas (with basic elaboration)

Rubric foundations

- **A-grade archetype** adherence + alignment + acceptable outcome OR planned small loss cleanly managed.
- **B-grade distinctions** sloppy win vs clean loss classifications differ—explicit matrix prevents rationalisation blobs.
- **C-grade remediation triggers** escalate to playbook freeze or playbook retirement candidacy—not eternal hope repeating.

Layered grading

- **Execution quality grading** fills path, improvisation absence, unintended partial exposure drift.
- **Deviation grading magnitude** micro vs cardinal sin classes with proportional responses.

Tracking

- **Process violation escalation ladder** patterned counts trigger standing reviews not mere annoyance shrug.
- **Missed-trade post-mortems** planned but aborted—signals fear or discipline or faulty alert architecture—different interventions.

Anatomy

- **Bad trade anatomies** root causes beyond "market wrong."
- **Good trade anatomies** strip survivorship halo: what was repeatable vs gratuitous adrenaline?

Insight loops

- **Mistake categorisation aggregates** trending cluster detection monthly.
- **Recurring-error detection heuristics** simple frequency threshold + thematic similarity lexical scan manual first.

Common pitfalls

- **Outcome overriding process** grading after one huge rogue bar—balance narrative carefully.

- **Analysis paralysis delaying next valid sample** obsessive rumination stagnates experiential pipeline.

Basic practice

- Weekly **distribution chart** grading counts—not only expectancy P&L.
- Monthly **mistake leaderboard** forcing top three addressed in backlog—not eleven shallow edits.

Phase — 12-trade-operations-lifecycle

Directory: `12-trade-operations-lifecycle/`

Source: `12-trade-operations-lifecycle/01-trade-operations-lifecycle.md`

Trade operations lifecycle — pre · during · post choreography

Scope

Stitch operating rhythm: **predictable preparation**, **controlled confrontation**, structured **retro-integration**. Loose improvisation increases variance—not “freedom”—without compensating virtuosity—which retail rarely sustainably possesses statistically.

Abbreviated directional contrast reminder:

```
trending environment playbook gating ≠ ranging environment gating indiscriminately
```

This lifecycle unit operationalises choreography more than ideology.

Areas (with basic elaboration)

Pre-trade cadence

- **Market preparation checklist** structural levels, liquidity notes, correlated markets snapshot—minimal viable not encyclopaedic daily unless style demands.
- **Economic calendar ingestion** tiered shocks: lethal vs nuisance vs misunderstood—pre-decide downgrade policies.
- **News / event overlays** asymmetric risk windows—when spread widens mechanically acceptable vs stand-down.
- **Volatility grids** hypothetical scenario planning (break up / down drift) aligning stop rationality—not fortune telling depth.
- **Invalidation maps** explicit structural breaks before entry emotional attachment solidifies entry bar.
- **Final handshake checklist** journaling template open, connectivity test, sizing cross-checked programmatically/manual double entry.

During-trade

- **Execution discipline** scripted sequences—no exploratory button pressing mid confusion.
- **Deviation interception cues** auditory / visual anchors when impulse rises—cheap psychology engineering.
- **Emotional micro-ownership labels** instantaneous note two words escalating later analysis not halting unnecessarily each tick paralysis.
- **Adaptive position-management** ONLY if pre-listed adaptive rules—not seat-of-pants improvisation new rule invention live.
- **Live adherence monitors** audible timer prompts review partial scale checkpoints if scripted.

Post-trade

- **Journal updates ingestion** ASAP fragility decay curve steep emotionally.
- **Screenshot archival pipeline** scripted folder naming taxonomy year/month/playbook subsets.
- **Mistake thematic extraction** bridging review engineering mistake tags.
- **Execution qualitative grading linkage** unify schema across systems.
- **Playbook grading** contextual environment tag alignment assessment.
- **Deviation taxonomy scoring** magnitude trends feed improvement backlog—not isolated shame artefacts.

Common pitfalls

- Skipping preparation on “obvious trending” euphoria days—often mean reversion ambush regimes statistically for some volatility clusters.
- **Post-trade journaling drift** collapsing into weekend binge reconstruction fiction distortion.

Basic practice

Print **lifecycle triptych cheat sheet**: three columns checklist sized wallet friendly card. Run **silent simulation passes** verbally narrating adherence without capital risk—to stress discipline architecture before rehearsal or live probing.

Phase — 13-market-context-engineering

Directory: `13-market-context-engineering/`

Source: `13-market-context-engineering/01-market-context-engineering.md`

Market context engineering — regime coherence

Scope

Environmental classification modulates playbook activation: correlations, overlaps, volatility & trend regimes, scheduled shocks. Mechanics ≠ environment—reuse invites silent degradation.

Structural reminder bridging psychology of strategic patience:

```
trending regime ≠ sideways regime playbook interchangeability blindness danger
```

Core ideas

- **Throttle matrix links context tags to playbook ON/OFF—not binary market tradable universally idea.**
- **Macro awareness coarse only unless macro is your authored edge** integrate calendar discipline still.

Areas (with basic elaboration)

Calendars & events

- **Economic calendar workflows** distinguishing surprise distribution tails vs scheduled ranges.
- **Earnings overlays** equities single-name exposures—gap redistribution risk windows.
- **Macro narrative scaffolding** heuristic risk-on/off without ideological capture—signals only not politics forum.

Time & overlaps

- **Session overlap potency mapping** correlating volatility upticks versus structural liquidity—not guaranteed formula learn instrument behaviour empirically personal logs.

Regime tagging

- **Volatility regime heuristics** IV rank analogues proxies depending market access document yours.
- **Trend regime heuristic labels** directional persistence vs compression metrics simplicity first refine later.

Cross-market

- **Event-risk classification ladders** escalate size-down tiers automatically written cold memorised verbally.
- **Correlation watch surfaces** co-moving shock baskets—prevent hidden duplication delta stacking narrative lying flat risk model.

Portfolio of contexts

- **Environment taxonomy powering gating filters** journaling tag alignment ensures stats meaningful stratification later analytics ROI.

Common pitfalls

- **Indicator identical parameters** oblivious contextual regime churn decay silent expectancy bleed months later puzzlement blame “market broken.”

Basic practice

Weekly **environment snapshot note** capturing five tags influencing next week playbook availability matrix adjustments allowed only via versioned playbook patch—not silent drift.

Phase — 14-psychology-engineering

Directory: `14-psychology-engineering/`

Source: `14-psychology-engineering/01-psychology-engineering.md`

Psychology engineering — mental operating system hygiene

Scope

Design **repeatable guardrails** for impulses, boredom, revenge patterns, tilt, and post-loss escalation. Psychological engineering treats mind-state as something you **instrument and rehearse**, not something you guilt yourself into magically fixing.

Individuals differ—none of these are universal cures—only experiments you log.

Core ideas

- Variance in markets interacts with unstable attention, fatigue, and stress. Assuming "always rational you" underestimates behavioural risk materially.

Areas (with basic elaboration)

Impulse regimes

- **FOMO interception** — Decide in advance what "too late" means for each playbook; scripted responses are smaller risk unit or no trade—not wider stops improvised hot.
- **Revenge-trade circuit breakers** — Escalating lockouts after patterned violations reset only via scheduled reviews, not improvised bargains while frustrated.
- **Tilt heuristic detection** — Simple cue list (anger spike, hurried clicking, journaling fields skipped) that triggers pause → flat → mandated review escalation.

Habits & boredom

- **Discipline anchors** — Stable preparation and shutdown rituals—even on flat days—so behavioural spine stays exercised.
- **Boredom-suppression substitutes** — Authorised list of absorbing non-market activities for moments when boredom drives random discretionary clicking.

Framing

- **Process-centred identity (light-touch)** — Preference for dependable rule adherence narratives over brittle "genius on good days" stories that euphorically inflate sizing after streaks.

Regulation modalities

- **Physiological resets** — Breath pacing, posture micro-adjustments, intentional short breaks away from screens; log what lowers urgency reliably for **you**.

Structural ownership

- **Routine artefacts** — Treat sleep steadiness and predictable anchors as upstream inputs to bandwidth—not mystical edge folklore.

Cognitive rehearsals

- **Loss acceptance rehearsal** — State permissible loss verbally or in journal before confirming risk—not to manifest optimism, but to reduce shock paralysis after normal adverse outcomes.

Common pitfalls

- Pure shame narratives that blame “weak character” while ignoring exhaustion, dehydration, or unrelated stress bleeding into keystrokes silently.

Basic practice

- Explore whether **emotion tags correlate with deviation tags** across your journals—pattern fodder only, not causal proof unless carefully studied.
- Write a numbered **tiered reset protocol**: pause → walk → mandatory remainder flat → mandated external review—with no live renegotiation of ordering.

Phase — 15-statistics-for-traders

Directory: `15-statistics-for-traders/`

Source: `15-statistics-for-traders/01-statistics-for-traders.md`

Statistics for traders — honest numeracy

Scope

Strengthen scepticism for claims and dashboards: dispersion, tails, expectancy sensitivity, and plausible ranges—not false precision dressed as rigor.

Treat probability language as a **humility tool**, not an ego shield.

Core ideas

- **Few trades hide huge uncertainty.** Intervals widen uncomfortably; ignoring that produces overconfidence sizing.
- **Monte Carlo (when used)** is sensitivity theatre unless model assumptions match your market and costs; illustrative, not prophecy.

Areas (with basic elaboration)

Probability & dependence

- **Probability primitives** feeding payoff thinking (conditional intuition, streak surprise).
- **Independence traps** trades often cluster by regime—serial correlation violates naive "fresh coin" fantasies unless your sample design proves independence.

Dispersion

- **Variance intuition** dominates short horizons more than expectancy means—fatigue and ruin couple to dispersion.
- **Standard deviation overlays** differentiate smooth-looking curves from lumpy empirical paths visually deceiving dashboards.

Expectancy realism

- **Expectancy formulations** must fold in costs (commissions/spread/swap/dividends) unless you consciously separate attribution layers—but never double-count fantasies vs live.
- **Median vs mean** skewed payoff streams differ—mean uplift from rare giant wins hides typical drawdown misery.

Drawdowns

- **Drawdown narratives** acknowledging absorbing barriers conceptually—even simple mental models discourage linear scaling fallacies leveraging "small predictable loss" stories.

Samples

- **Sample size scepticism** rare playbook tags starve inference—explicitly segregate qualitative vs quantitative conclusions.
- **Confidence humility heuristics** prefer ranges ("expectancy plausible band") over single scalars shouted as truths.

Distributions

- **Skew awareness** payoff shapes dictate journal emphasis (tail hunting vs grinder styles).

Simulation

- **Monte Carlo awareness** mandates listing omitted dynamics: regime shifts, liquidity gaps, behavioural shutdowns—you model partial worlds.

Common pitfalls

- **Sharpe/pseudo-Sharpe** on tiny samples as marketing-grade self-affirmation.
- Confusing curve smoothness online with **risk of ruin** practicality under leverage bursts.

Basic practice

For any headline KPI you glance weekly, add a one-line note: **“This metric lies if _____.”** Rotate the blank weekly as a scepticism ritual.

Phase — 16-portfolio-construction

Directory: `16-portfolio-construction/`

Source: `16-portfolio-construction/01-portfolio-construction-awareness.md`

Portfolio construction — multi-surface coherence

Scope

Apply when multiple sleeves coexist (cash equities vs FX vs synthetic products, multicurrency exposures, thematic overlap across accounts). Aggregation reveals **overlap** journaling single-symbol rows often hides until shocks align exposures.

Treat portfolio posture as concentration of **risk and narrative**, not a long list of different tickers.

For tax overlays in Germany—or cross-border bookkeeping—coordinate with advisers; this unit stays operational and thematic, not prescriptive legally.

Core ideas

- **Correlation beats symbol count.** Diversifying names without diversifying sensitivities concentrates risk quietly.
- **Reporting currency versus risk currency can diverge.** FX overlays matter for shocks, bookkeeping, and margin surprises even when you think “mostly domestic.”

Areas (with basic elaboration)

Diversification & correlation

- **Naive diversification** — Many positions can still behave like **one macro bet** sharing sector earnings cycles, commodities inputs, liquidity regime, or index beta.
- **Correlation stewardship** — Revisit thematic overlap after regimes shift—not a one-time spreadsheet exercise.

Exposures & netting

- **Gross vs net intuition** — Netting correlated exposures thoughtfully; “hedged” can still stack correlated losers if correlations spike in stress.

Sizing between sleeves

- **Concentration budgeting** — Thematic caps that span brokers and wrappers; consolidate views so fragmentation does not hide stacked bets.

Allocation

- **Capital allocation ladders** — Explicit slices for exploratory research sleeves vs validated playbooks, with gates for promotion defined elsewhere (risk, deployment, journaling evidence).

Weighting philosophies

- **Conviction-skew versus risk-parity-lite** styles—pick deliberately and document biases so hindsight cannot silently rewrite motives.

Cross exposures

- **Sector / thematic overlays** — Unified tagging across sleeves (for example overlapping supply-chain exposure in multiple instruments); optional shock tabletops deepen intuition without requiring heavy quant early.

FX

- **FX exposure bookkeeping** — Track where margin, reporting denomination, and real economic sensitivities diverge—align with archival and adviser workflows cautiously where tax touches.

Drawdown coupling

- **Portfolio-level drawdown triggers** — Activate throttles on aggregate curves, not only isolated playbook KPIs cheering while another sleeve silently bleeds in shared macro moves.

Common pitfalls

- Stacking leveraged synthetic exposure atop correlated cash equities without realising **doubled thematic beta** until a shared shock proves it.

Basic practice

- Maintain an **overlap map**: top three thematic concentrations across all holdings—even a manual table refreshed at review tempo beats drifting blind diversification stories.

Phase — 17-trading-business-operations

Directory: `17-trading-business-operations/`

Source: **`17-trading-business-operations/01-trading-business-operations.md`**

Trading business operations — treat longevity like an operations job

Scope

Serious trading over years benefits from **cheap operational hygiene**: artefacts you can revisit, KPIs you define up front, and rehearsals so broker or tax deadlines do not invent panic work.

Tailor depth to scale—solo retail rarely needs enterprise paperwork cosplay; you need **few durable rails** actually used weekly.

Core ideas

- **Operational debt compounds** like technical debt: postpone exports, naming, reconciliation once, pay interest as confusion for months.

Areas (with basic elaboration)

Ownership & artefacts

- **Documentation ownership matrix** — who maintains playbooks vs broker notes vs archive folders; revisit when tooling changes.

Procedure layers

- **Trading SOPs** layered by severity: outage flatten path, reconciliation path, journaling minimum viable path—short documents beat fantasy manuals.

Archives

- **Trade archive integrity & versioning** — folder taxonomy by year/quarter/playbook/broker; optional checksums purely for tamper-awareness discipline.

Cadence

- **Review cadence** micro (daily adherence tick), macro (weekly pattern scan), quarterly strategic (playbook deprecation candidates).

KPI dashboards

- **KPI definitions beyond headline P&L** — adherence rate, mistake recurrence, median slippage, fee drag rows, downtime days—pick a small invariant set first.

Compliance & tax rehearsal

- **Compliance workflow rehearsals** — dry-run exporting everything your adviser asked for once, note gaps calmly not under deadline glare.
- **Tax preparation workflow staging** integrated with archival discipline (consult professionals for substance).

Annual rhythm

- **Annual reporting rehearsals** verifying field coverage vs prior filings pattern only as advisor-directed checklist scaffolding.

Common pitfalls

- Building dashboards nobody refreshes → **entropy returns** silently.
- **Fragmented artefacts** across three brokers without consolidated exposure view—even a monthly CSV merge suffices early.

Basic practice

Create **three living documents**: (1) SOP index, (2) KPI glossary, (3) archive map. Refresh them periodically at your review tempo—expire anything obsolete instead of hoarding scaffolding.

Phase — 18-germany-trading-operations

Directory: `18-germany-trading-operations/`

Source: **`18-germany-trading-operations/01-germany-trading-operations-checklists.md`**

Germany trading operations — checklists requiring professional verification

Scope

Organise operational habits when your posture touches **German bookkeeping and reporting workflows**: exports, reconciliation, consolidation, adviser handoffs.

Tax law, classifications, exemptions, allowances, deadlines, forms, annexes change. Only an adviser + official sources anchor decisions.

This file is deliberately **generic study scaffolding**.

Disclaimer

Nothing here is tax or legal advice. Keep a dated log when rules or interpretations shift.

Core ideas

- Evidence you can reproduce beats memory when platforms change exports.
- Separation of duties: **you own execution logs**, **counsel owns filing correctness** mapping—coordinate explicitly.

Areas (with basic elaboration)

Themes to verify yearly

- **Capital gains thematic awareness** — verify holding-period logic, netting concepts, exemptions, wrappers (if applicable) with counsel—not generic forums.
- **Broker reporting ingestion** — which exports your adviser expects (CSV schemas, PDF statements, withholdings summaries).

Workflow artefacts

- **Typical artefacts checklist** confirmations, dividend vouchers, withholding lines, FX conversion notes—as prompts for your accountant, not a universal list.

Retention

- **Document retention discipline** immutable copies per tax-relevant bundle; annotate broker UI version/date when exporting.

Classification

- **Private vs habitual commercial framing** belongs with counsel-only discussion—avoid DIY threshold mythology.

Collaboration loops

- **Accountant workflow** — agreed timing (monthly summaries vs annual dump), ticketing questions, clarification cadence preventing December panic choke.

Operational schema

- **Single-source schema** bridging internal blotter ↔ broker ↔ adviser worksheet fields—prevent column mismatch rework.

Evidence hygiene

- **Evidence bundles** zipped by fiscal year subsets optional—keep human-readable filenames not opaque hashes exclusively.

Multi-broker

- **Cross-broker consolidation** manual merge discipline until tooling exists—weekly micro reconciliation lowers annual merge shock.

Rehearsal

- **Pre-submission rehearsal** dry-run completeness scan: missing months, orphaned symbols unexplained deltas flagged early.

Common pitfalls

- Mixing speculative forum posts with authoritative filing narratives without adviser sign-off.

Basic practice

Perform **cold export rehearsals** on quiet days: regenerate the artefacts bundle your adviser expects, validate naming/versioning rituals, patch gaps early—stress later should not be the first test.

Phase — 19-system-evolution

Directory: `19-system-evolution/`

Source: `19-system-evolution/01-system-evolution.md`

System evolution — change control for discretionary & hybrid systems

Scope

Regimes drift, liquidity changes, brokerage behaviour shifts, journaling habits mature. Playing rules that secretly morph produces statistics you cannot interpret.

Controlled evolution blends **experimentation hygiene**, **explicit versions**, deliberate **rollback**, and sometimes **strategy retirement**.

Core ideas

- **Invisible edits scramble comparability.** You need version identifiers on playbooks aligned with journaling tags—or reviews compare apples from different trees.
- **Retiring losers is success.** Carrying sentimental systems inflates backlog noise and wastes attention.

Areas (with basic elaboration)

Detection

- **Degradation heuristics** — Recurring softness in payoff or adherence within a tagged playbook—not one cold streak—informs investigation.

Baselines

- **Benchmark upkeep** naive comparators ageing unfairly versus macro eras; revisit baseline fairness during major reviews.

Change discipline

- **Freeze windows / lab segregation** exploratory draft rules forbidden from silently mixing into live tagged trades without versioning cutover.

Governance

- **Parameter authority rituals** documenting who adjusts what—even solo personas use scheduled calm reviews—not live whim.

Evidence trail

- **Change logs bridging intent outcomes** — Maintain short entries linking **intent → measurement → verdict** whenever rules move.

Versioning rollback

- **Semantic labels** — Simple playbook version stamps aligned with journal metadata fields.

Common pitfalls

Micro-tweaking after noisy samples—optimisation disguised as pattern recognition.

Basic practice

Append a playbook **changelog**: even one sentence per revision linking rationale to KPI shifts worth caring about at your review tempo.

Phase — 20-deployment-framework

Directory: `20-deployment-framework/`

Source: **`20-deployment-framework/01-deployment-framework.md`**

Deployment framework — guarded live ramps

Scope

Bring systems from notebooks, backtests, and paper paths into live capital gradually using **gates, observation, escalation limits, rollback, and orderly retirement**. The goal is to avoid flipping from "research success story" straight into full conviction exposure under enthusiasm.

Bridging mnemonic (illustrative, not doctrine):

```
idea → research plus bias ledger
    → replay or aligned paper journaling
    → small live probing with hard caps
    → monitored guarded scale ladders
    → bounded adjustments plus retirement pathways
```

Core ideas

- **Staged sizing** absorbs execution quirks and psyche adaptation costs before larger fractions of capital participate.
- **Forward reality inserts friction** (slips, interruptions, emotional variance). Gates exist to uncover those issues early, while downside remains contained.

Areas (with basic elaboration)

Evidence thresholds

- Write what qualifies as sufficient evidence before widening units: journaling depth, playbook version discipline, reproducibility signals from replay/paper, and an explicit realism ledger from prior research.

Deployment checklists

- One consolidated checklist that ties together risk sheet sizing, playbook version stamp, journaling schema readiness, outage/flatten rehearsals, kill-switch realism, plus adviser/tax artefacts if applicable.

Scaling ladders

- Define stair-steps tied to degrade signals staying quiet across agreed observation spans. "Observe" means review artefacts, not hope.

Small-size probing

- Keep earliest live fractions intentionally modest versus eventual target envelope so measurement survives mistakes.

Degrade alerting

- Watch expectancy softness, payoff distribution drift (with regime awareness), journaling adherence slump, recurring execution failures, recurring deviation clusters.

Kill switches

- Decide what halts automation or discretionary participation automatically versus manually—and keep redundant manual kills when platform automation cannot be trusted.

Strategic halt vs tactical halt

- Distinguish **intraday/session circuits** from **strategy retirement**. Retirement belongs in cold writing with criteria—not improvised discouragement loops.

Rollback

- Shrinking capital units plus restoring a coherent prior playbook version—not a muddy hybrid accidentally mixing eras.

Common pitfalls

- Scaling after unusually lucky early samples (variance masquerading as validation).

Basic practice

- Maintain a reusable **deployment worksheet** filling: evidence summary, promotion criteria, degrade signals checklist, rollback trigger list, retirement sketch.

Phase — 21-full-trading-operating-system-lab

Directory: `21-full-trading-operating-system-lab/`

Source: `21-full-trading-operating-system-lab/01-full-trading-operating-system-lab.md`

Full trading operating system lab — integrating the stack

Scope

Single capstone pass that touches **risk, playbook, execution, journaling, reviews, archiving, KPIs**, plus **deployment/evolution stubs** and **adviser-facing question stubs** when filings might apply.

Thin slices per pillar are fine—the win is coherence across artefacts, not one perfect subsection.

Goal: practise a **Trading OS**, not collect indicator lore.

Core ideas

- Strong fragments degrade without rails: **version stamps, review ingestion paths, repeatable exports**.
- The lab succeeds when it **shows gaps honestly** rather than ticking boxes theatrically.

Integrated loop (adapt your tools)

```
broker realism → setup tag + playbook version
→ thesis + falsifiers → execution choreography
→ risk conformance → journal + artefacts
→ graded review → KPI delta (minimal definition set pinned)
→ backlog patches linked to playbook version
→ filing/evidence stubs (substance verified with advisers/official texts)
→ evolution changelog append
```

Pillar checks

Broker / custody + validation drills

- Blotter ↔ statement artefacts reconcile—or flag deltas—and attach Broker Validation artefacts (spread sample, latency feel-check, export parse proof, outage drill note, hotkey map review).

Playbook fidelity

- Version ID visible where rules live aligns with journaling tags; deviation tags map cleanly.

Execution discipline

- Written default order path plus explicit fallback behaviours for rejects, disconnects, and sharp spread widening.

Risk containment

- Sizing cites the active risk sheet; rehearse verbally how preset **circuits/throttles** would operate under clustered losses without hot improvisation.

Journals → reviews

- Sample review rubric scores **plan adherence beside outcome**, not headline P&L alone.

KPI scaffolding

- Reuse one **small KPI definition list** everywhere so renaming cannot hide drift later.

Archiving / ops

- Proof-of-concept artefacts follow your naming scheme so retrieval is mechanical.

Lab bundle (minimum artefacts)

Produce one integrated bundle—even shallowly—plus **an omissions list**.

Suggested entries:

1. Journal excerpt using your schema faithfully.
2. Playbook excerpt with version footer plus changelog snippet.
3. Risk sheet excerpt tying to illustrative sizing maths.
4. Execution checklist excerpt with failure behaviours.
5. Review rubric excerpt with adherence-centric scoring sketch.
6. Archive naming proof aligning with taxonomy.
7. Deployment worksheet excerpt (promotion/degrade/rollback prompts).
8. Counsel-facing question stubs only—**never substitute this file for tax/legal conclusions**.

Common pitfalls

- Declaring completeness while hiding unanswered questions.

Basic practice

After assembly log **five fragilities**, each linking to exactly one backlog item with playbook version metadata.
